Cavity-induced persistent current in a π loop

Model. We consider a superconducting (SC) loop of L sites closed by an interacting quantum dot (site 0) and threaded by the magnetic flux of a single lumped LC mode [Fig. 1(a)]. The full microscopic Hamiltonian is

$$H = \hbar\omega\, a^\dagger a - t\sum_{r=1}^{L-2}\sum_{\sigma}\left(c^\dagger_{r+1\sigma}c_{r\sigma} + \text{h.c.}\right) + \Delta\sum_{r=1}^{L-1}\left(c^\dagger_{r\uparrow}c^\dagger_{r\downarrow} + \right.$$
$$+ \epsilon_d\, n_0 + U n_{0\uparrow}n_{0\downarrow} - t_J\sum_{\sigma}\left(c^\dagger_{0\sigma}c_{L-1\,\sigma} + \text{h.c.}\right)$$
$$- t_J\sum_{\sigma}\left(e^{i\varphi_{\rm cl}}\, e^{ig(a+a^\dagger)}\, c^\dagger_{1\sigma}c_{0\sigma} + \text{h.c.}\right). \qquad (1)$$

The first line is the resonator and an s-wave BCS chain (hopping t, pairing Δ); the second line is the Anderson impurity (level $\epsilon_d = -U/2$, on-site repulsion U) tunnel-coupled (t_J) to the two ends of the chain; the third line carries the total Peierls phase of the loop, which by a gauge choice is placed on a single bond. The vector potential of the mode enters through the gauge-invariant displacement operator $D = e^{ig(a+a^\dagger)}$, so diamagnetic (A^2) terms are included to all orders; $g = 2\pi\Phi_{\rm zpf}/(h/e)$ is the electron phase generated by the zero-point flux of the mode, and $\varphi_{\rm cl}$ is a static external flux. An electron phase $\varphi = \pi$ corresponds to one superconducting flux quantum $\Phi_0 = h/2e$. For a lumped resonator of impedance Z, galvanically sharing its inductance with the loop, $(2g)^2 = \pi Z/R_Q$ with $R_Q = h/4e^2$. Unless stated otherwise we take $t = 1$, $\Delta = t$, $t_J = 0.5t$, $L = 14$ ($L \gg \xi$), $\hbar\omega = 7.6\times10^{-4}t$ and *no* external flux, $\varphi_{\rm cl} \to 0^+$ (a seed $\varphi_{\rm cl} = 10^{-3}$ is used to select one of the degenerate states). Two auxiliary versions are also used: a weak link ($U = 0$, dot replaced by a bond t_J) biased by $\Phi_0/2$, and the short ring with a Zeeman-polarized dot of Ref. [1]. We write Eq. (1) compactly as $H = \hbar\omega a^\dagger a + H_{\rm ring}[\varphi_{\rm cl} + g(a + a^\dagger)]$.

The only property of $H_{\rm ring}$ that enters below is its ground-state energy at fixed phase,

$$E(\varphi) = \sum_{m\geq 0} E_m \cos m\varphi. \qquad (2)$$

Even harmonics ($m = 2, 4, \dots$) describe Cooper-pair tunnelling ($h/2e$ periodicity). The $m = 1$ term is the single-electron (h/e) persistent current of a mesoscopic ring [2] and is suppressed as $e^{-L/\xi}$. A 0-junction has $E_2 < 0$ and a π-junction has $E_2 > 0$. When a 0-junction is biased by $\Phi_0/2$, $\varphi_{\rm cl} = \pi/2$ maps $E_2\cos 2\varphi \to -E_2\cos 2\varphi$. The two routes to a π loop are therefore equivalent up to the odd harmonics, which turn into an explicit symmetry-breaking term $-E_1\sin\varphi$. In contrast, an intrinsic π-junction keeps $E(\varphi) = E(-\varphi)$ exactly, since $H_{\rm ring}[0]$ is real (time-reversal symmetry).

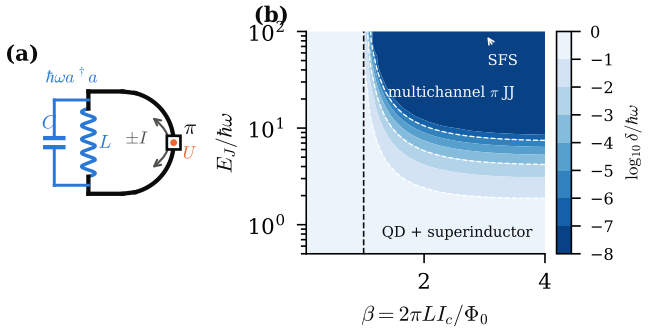


FIG. 1. (a) A superconducting loop closed by a π junction (here an interacting dot with charging energy U) shares its inductance L with an LC mode; the two degenerate states carry opposite persistent currents $\pm I$. (b) Tunnel splitting δ between the two current-carrying states in the plane of the screening parameter β and the quantum parameter $E_J/\hbar\omega$ [Eq. (3) with the DMRG $E(\varphi)$ of the $U = 2t$ junction]. White lines: $\delta/\hbar\omega = 10^{-6}, 10^{-3}, 10^{-1}$; dashed: classical threshold $\beta = 1$. Labels: typical regimes of the platforms discussed in the text.

Adiabatic reduction and Landau theory. For $\hbar\omega$ well below the Andreev gap E_A, the electrons follow the flux adiabatically. To leading order in $\hbar\omega/E_A$, Eq. (1) reduces to a single-particle problem for the dimensionless flux $X = a + a^\dagger$,

$$H_{\rm BO} = \frac{\hbar\omega}{4}\left(P^2 + X^2\right) + E(\varphi_{\rm cl} + gX), \qquad [X, P] = 2i. \qquad (3)$$

Equation (3) is the Hamiltonian of an rf-SQUID [3] whose inductive energy is supplied by the mode. Writing it in the Cooper-pair phase $\theta = 2gX$ gives $4E_C n^2 + \frac{1}{2}E_L\theta^2 + E(\theta/2)$ with $E_L = \hbar\omega/8g^2$ and $E_C = \hbar\omega g^2$. A static field ($\langle a\rangle \neq 0$), i.e. a photon condensate carrying a persistent current $I = -(2e/\hbar)\partial_\varphi E$, appears classically when the orbital response of the ring at the symmetric point is paramagnetic and exceeds the stiffness of the mode,

$$\beta \equiv \frac{2g^2\,|E''(\varphi_0)|}{\hbar\omega} > 1, \qquad E''(\varphi_0) < 0. \qquad (4)$$

For a pure π-junction, $\beta = 8g^2E_J/\hbar\omega = 2\pi LI_c/\Phi_0$, the familiar screening parameter of the loop [4, 5]. Equation (4) is the single-loop form of the photon-condensation criterion of Andolina *et al.* [6]: the no-go theorem for a uniform cavity field [7, 8] does not apply, because the mode couples to the flux (curl of $\mathbf{A}$), and the instability requires a *paramagnetic* orbital response, which here comes from the π shift. Above threshold, the Landau free energy $\mathcal{F}(X) = \hbar\omega X^2/4 + E(gX)$ has two degenerate minima $\pm X^\star$ carrying opposite currents $\pm I^\star$.

Two remarks fix the physical content of Eq. (4). First, for a loop with its own geometric inductance L_r and

mutual inductance $M \le \sqrt{L_r L_c}$ to the resonator, minimizing the static energy over the resonator flux gives $\Phi_r^2/2L_r$: the static threshold is set by the total loop inductance, and Eq. (1) is the $k = M/\sqrt{L_r L_c} \to 1$ (galvanic) limit. Second, zero-point fluctuations of the mode do not drive the transition; they oppose it. At finite $E_J/\hbar\omega$ the ground state of Eq. (3) is the symmetric superposition of the two current-carrying states, split from its partner by a tunnelling gap $\delta \sim \hbar\omega\, e^{-S}$ with an instanton action $S \propto (E_J/\hbar\omega)\, s(\beta)$. A sharp quantum phase transition with mean-field exponents is recovered in the scaling limit $E_J/\hbar\omega \to \infty$ at fixed β, in which $E_C/E_J = \beta(\hbar\omega)^2/(8E_J^2) \to 0$. This is the analogue of the classical-oscillator limit of the quantum Rabi model [9].

Methods. We solve Eq. (1) without approximation by DMRG (TeNPy [10]) on the full electron–photon Hilbert space. The ring is stored in folded order next to a bosonic site (Fock cutoff $N_{\rm ph}$). Fermion parity and S^z are conserved, and the electron–photon vertex is a three-site operator $D\, c_{1\sigma}^\dagger c_{0\sigma}$. Benchmarks against exact BdG energies of the non-interacting ring at $g = 0$ agree to 10^{-14} at full bond dimension ($\chi = 1024$). In production ($\chi = 256$–320, $N_{\rm ph} = 30$–50, discarded Fock weight $< 10^{-7}$) the ground-state energy error is $\lesssim 3\times10^{-6}t$. For a 4-site ring with a photon, DMRG reproduces exact diagonalization and the identity $\langle I\rangle = \hbar\omega\langle X\rangle/2g$ to 10^{-7}. Excited states are obtained by DMRG orthogonal to the ground state. We compare with (i) the self-consistent electron×photon product state (mean field, MF), and (ii) the adiabatic Hamiltonian (3), with $E(\varphi)$ from BdG or from DMRG at fixed φ, diagonalized exactly.

Choice of regime and dimensionless variables. The microscopic model (1) contains the energy scales t, Δ, t_J, U of the junction and $\hbar\omega, g$ of the mode. Three dimensionless ratios control the physics. (i) U/Δ at fixed Γ/Δ (here $\Gamma = 2t_J^2/t = \Delta/2$) sets the magnetic state of the dot and hence the sign of the Josephson coupling: the loop is a π loop only for $U > U_c$. (ii) The coupling g is fixed by the impedance of the mode alone, $g^2 = \pi Z/4R_Q$, i.e. $g = 0.1$–0.4 for $Z = 0.08$–$1.3\,\mathrm{k\Omega}$, the range of superinducting resonators [11, 12]. (iii) $E_J/\hbar\omega$, with $E_J \equiv E_2 > 0$ the Josephson energy of the π junction, decides whether the transition is sharp: the two current states tunnel with $\delta \sim \hbar\omega e^{-S}$, $S \propto E_J/\hbar\omega$. We take $\hbar\omega = E_J(U = 2\Delta)/10 = 7.39\times10^{-4}\Delta$: for a dot junction with $I_c^\pi \simeq 10\,\mathrm{nA}$ ($E_J/h \simeq 5\,\mathrm{GHz}$) this is a 500 MHz mode, as realized by a loop of superinductance $L \simeq \Phi_0/2\pi I_c \simeq 30\,\mu\mathrm{H}$, i.e. a fluxonium-type circuit [11]. With this choice the classical threshold $\beta = 8g^2E_J(U)/\hbar\omega = 1$ lies at $g = 0.11$ for $U = 2\Delta$ and sweeps across the experimentally accessible window $g \le 0.4$ as U is varied from U_c to 8Δ, while $E_J/\hbar\omega$ decreases from 15 to 2, so that the same diagram contains the sharp transition and its quantum washing-out. We present it in the plane $(U/\Delta, g)$; the right axis of Fig. 2 converts g to Z/R_Q. The ring has $L = 10$ sites ($L \gg \xi \simeq t/\Delta$): the π-sector Josephson energy is converged to 0.2% between $L = 8$ and $L = 14$ ($E_2 = 7.403, 7.392, 7.391, 7.391 \times 10^{-3}\Delta$ for $L = 8, 10, 12, 14$) and the single-electron harmonic is $|E_1|/E_2 \le 2\%$.

Gauge-invariant current operator. Since the Peierls phase of the loop is placed on the dot–ring bond [Eq. (1)], the persistent current is the derivative of H with respect to the total flux, $I = -(2e/\hbar)\,\partial H/\partial\varphi_{\rm cl}$,

$$I = \frac{2e}{\hbar}\, i\, t_J \sum_\sigma \left(e^{i\varphi_{\rm cl}} e^{ig(a+a^\dagger)} c_{1\sigma}^\dagger c_{0\sigma} - \text{h.c.} \right) \qquad (5)$$

(below we quote I in units of $2e\Delta/\hbar$). This is the particle current through the dot–ring bond, dressed by the displacement operator of the mode: the combination $e^{i(\varphi_{\rm cl}+gX)}c_1^\dagger c_0$ is invariant under the simultaneous gauge transformation of the electrons and of the vector potential, and its expectation value does not change if the Peierls phase is distributed along the ring with the pairing amplitude transformed covariantly [13]. Because the dot carries no pairing term, the particle number of site 0 is conserved by H and the current entering the dot from site $L-1$ equals the current (5) leaving it; the same is not true of the bonds inside the BCS chain, where the fixed-phase pairing term acts as a condensate reservoir [13]. Equation (5) obeys an exact sum rule. With $X = a + a^\dagger$ and $P = -i(a - a^\dagger)$, $[H, P] = i\hbar\omega X - 2ig\,(\hbar/2e)I$, so that in any eigenstate

$$\langle I\rangle = \frac{2e}{\hbar}\,\frac{\hbar\omega}{2g}\,\langle a + a^\dagger\rangle : \qquad (6)$$

the persistent current *is* the static displacement of the mode (the dc flux it generates, $\Phi = LI$). We evaluate both sides independently in DMRG and use their difference as a convergence measure: $\langle I\rangle$ is a local fermionic observable, $\langle X\rangle$ a photonic one, and only an eigenstate satisfies Eq. (6).

DMRG protocol. At each (U, g) we obtain the ground state of Eq. (1) on the full electron–photon Hilbert space by two-site DMRG with bond dimension $\chi = 300$ (truncation error $< 2\times10^{-8}$), a photon cutoff $N_{\rm ph} = 50$–60 chosen such that the population of the last Fock state is $< 10^{-7}$, and a seed $\varphi_{\rm cl} = 10^{-3}$ that selects one of the two degenerate current states. The grid is $g = 0.02$–0.40 in steps of 0.01 (0.02 above 0.3) and $U/\Delta = 0, 0.5, 0.75, 0.85, 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8,$ 427 ground states in all (about 20 CPU hours). Fermion parity and S^z are conserved; for $U \le \Delta$ both parity sectors are computed and the lower one is taken, which locates the 0–π boundary of the *loop* without any assumption. The two current-carrying states are split only by the exponentially small δ and by the seed, $2I^\star\varphi_{\rm cl} \sim 10^{-5}\Delta$, so that the energy landscape between them is flat and local DMRG updates drift towards the polarized state over tens of sweeps. We therefore use two independent initial states at every point and keep the lower-energy

result: (A) the converged state of the neighbouring g, which approaches the transition from the vacuum side, and (C) a product state with the mode in a coherent state at the classical flux, converged first with a large seed ($\varphi_{\rm cl} = 10^{-2}$, then 3×10^{-3}) and then relaxed at 10^{-3}, which approaches it from the polarized side. Where the two differ, the energy decides whenever the difference exceeds $3\times10^{-7}\Delta$; the 47 points (11% of the grid, all inside the crossover windows discussed below) where it does not are reported as brackets [Fig. 4(b)]. The sum rule (6) is satisfied to better than $10^{-3}I_c$ at every point with $g \geq 0.03$ (median $10^{-4}I_c$) [Fig. 4(a)]. Increasing χ to 400 changes the energies by $< 10^{-8}\Delta$ and the current by $< 0.1\%$ except at the crossover point; raising $N_{\rm ph}$ to 80 changes the energy by $< 10^{-9}\Delta$; and reducing the seed to 10^{-4} leaves the current above threshold unchanged to 0.4% while the current below threshold scales down by the factor 0.09 [Fig. 4(c)]: the current is spontaneous.

Effective circuit model. When $\hbar\omega$ is small compared to the Andreev gap E_A of the junction ($\hbar\omega/E_A \simeq 10^{-3}$ here), the fermions can be integrated out adiabatically: the mode moves in the exact ground-state energy of the ring at fixed phase and fixed parity, $H_{\rm eff}^{(s)} = \hbar\omega a^\dagger a + E_s(\varphi_{\rm cl} + gX; U)$, $s = {\rm even}, {\rm odd}$ [Eq. (3)], an rf-SQUID Hamiltonian whose inductive energy is supplied by the mode; parity conservation forbids any mixing of the two surfaces, and the ground state is the lower of the two sector ground states. The matching of parameters is unambiguous: $E_s(\varphi; U)$ is computed by DMRG at $g = 0$. The doublet surface is represented by four harmonics (nine phases); the singlet surface has a cusp at $\varphi = \pi/2$ (a level crossing inside the even sector) and is represented by a spline through 25 phases. $H_{\rm eff}$ is then diagonalized exactly in a Fock space of 200 states, with the same seed. Its current is $I_{\rm eff} = -(2e/\hbar)\partial_\varphi E_s$, the classical current of the junction averaged over the flux distribution of the mode; the omitted terms are the non-adiabatic (Berry) corrections of relative order $\hbar\omega/E_A$. The mean-field theory of Ref. [1], generalized to the interacting dot (Hartree–Fock BdG electrons × photon product state, swept upwards in g with the same seed), is evaluated on the same grid.

Phase diagram. Figure 2 shows $|\langle I\rangle|/I_c(U)$, with $I_c(U)$ the critical current of the π junction at $g = 0$, from the full DMRG (a), the effective circuit model (b) and mean field (c); Fig. 3 shows cuts. The exact diagram has three regions. (i) For $U < U_c(g)$ the loop is a 0 loop: the singlet is the ground state at all g, the orbital response is diamagnetic and the mode is not softened but stiffened, $\omega \to \omega\sqrt{1 + 2g^2|E''|/\hbar\omega}$, i.e. squeezed ($\langle a^\dagger a\rangle = 1.9$ virtual photons at $g = 0.4$, $|\langle I\rangle| < 10^{-2}I_c$ and linear in the seed). The boundary is not the zero-flux 0–π crossing of the junction, $U_c^0 = 0.94\Delta$: the doublet gains the condensation energy of the π loop (up to $2E_J$), so that $U_c(g)$ decreases from 0.94Δ at $g \to 0$ to 0.86Δ at $g = 0.4$ [at $U = 0.85\Delta$, $E_{\rm odd} - E_{\rm even}$ falls from $2.3\times10^{-2}\Delta$ to $9\times10^{-4}\Delta$]. The flux of the mode thus drives a parity switch of the junction, and in the window $0.83\Delta < U < 0.94\Delta$, where the global minimum of $E(\varphi)$ already belongs to the doublet, the switch occurs at a finite coupling ($g \simeq 0.3$ at $U = 0.9\Delta$ in the effective model). (ii) For $U > U_c$ and $g > g_c(U)$ the mode condenses: at $U = 2\Delta$ the photon number rises from 0.5 at $g = 0.11$ to 5.7 at $g = 0.13$ and 7.3 at $g = 0.14$, where the current reaches $0.96\,I_c$; it then decreases as $I \simeq (2e/\hbar)\hbar\omega\varphi^\star/2g^2$, because the condensate flux saturates at $\Phi_0/2$ while the inductance $\propto g^2/\hbar\omega$ grows [Eq. (6)]. The threshold $g_c(U)$ (first grid point with $|\langle I\rangle| > 0.5I_c$) rises from 0.10 at $U = \Delta$ to 0.13, 0.19, 0.24 and 0.32 at $U = 2, 4, 5, 6\Delta$, following the decrease of $E_J(U)$. It lies systematically *above* the classical line $\beta = 1$ (dash-dotted), by $\Delta g = 0.02$ at $U = 2\Delta$, 0.04 at 4Δ and 0.12 at 6Δ: the zero-point motion of the flux opposes the condensation, and the shift grows as $E_J/\hbar\omega$ decreases from 10 to 3.2. At the same time the transition broadens: the width of the crossover in g grows from one grid step at $U \leq 3\Delta$ to 0.05 at $U = 5\Delta$ and 0.08 at $U = 6\Delta$, and the maximal current drops from $0.97I_c$ ($U = \Delta$) to 0.90, 0.81 and $0.61I_c$ ($U = 4, 5, 6\Delta$). For $U = 8\Delta$ ($E_J/\hbar\omega = 2.1$) no ordered phase is left in the window $g \leq 0.4$ although β reaches 2.7: there the ground state is the cat state with $\delta \simeq \hbar\omega$ and $\langle I\rangle \leq 0.04I_c$, proportional to the seed. This is the finite-$E_J/\hbar\omega$ remnant of the transition, which becomes sharp only in the limit $E_J/\hbar\omega \to \infty$ at fixed β (Fig. 1).

Comparison with the effective model and with mean field. Above threshold the full DMRG and the effective circuit model agree to 0.2% of I_c on average and to 2.4% at worst over the whole ordered region, and to 0.1 in the photon number (the odd-sector energies agree to $10^{-6}\Delta$ at all g), which validates the adiabatic elimination of the fermions at $\hbar\omega/E_A \sim 10^{-3}$. The only points where they differ by more are the bracket points inside the crossover windows, where the exact ground state is a partially polarized cat and the DMRG from the two initial states brackets the effective-model value from both sides [vertical bars in Fig. 3(a)]. This is a statement about the convergence of a variational method on a flat energy landscape, not about the physics: the energy difference between the two starts is $< 3\times10^{-7}\Delta$, while the effective model, which resolves it exactly, interpolates smoothly between them. Mean field locates $g_c(U)$ to within the grid step for $U \leq 5\Delta$ (0.27 versus 0.32 at 6Δ) but underestimates the current by 10–20% (0.86 versus $0.96\,I_c$ at $U = 2\Delta$; 0.72 versus $0.90\,I_c$ at 4Δ) because the product state renormalizes the junction bond by $|\langle D\rangle| < 1$; it overestimates the photon number at large g (5.2 versus 3.7 at $U = 2\Delta$, $g = 0.4$), where the exact flux distribution is a displaced squeezed state rather than a coherent state; it predicts an ordered phase with $0.6I_c$ at $U = 7$–8Δ that the exact solution does not have, because a product state cannot describe the tunnelling between the two current

directions that restores the symmetry when $E_J \lesssim 3\hbar\omega$; and, since its Hartree–Fock threshold $U_c^{\rm MF} = 0.82\Delta$ replaces the Kramers doublet by a static moment, it orders the loop at $U = 0.85\Delta$ with $0.95 I_c$ where the exact ground state is the singlet 0 loop.

Spontaneous persistent current in the microscopic model. Figure 5 contains the central result. Panel (a) follows the ground state of Eq. (1) at $U = 2t$ as the coupling g is increased. For $g < g_c \simeq 0.12$ the mode stays close to its vacuum ($\langle a^\dagger a\rangle \lesssim 1$) and the current is the linear response to the infinitesimal seed flux, $|\langle I\rangle| \lesssim 10^{-4} et/\hbar$. At $g \simeq g_c$ the photon number jumps to $\langle a^\dagger a\rangle \simeq 8$ and a persistent current $|\langle I\rangle| \simeq 1.4\times10^{-2} et/\hbar \simeq 0.93\, I_c$ appears. Reducing the seed by two orders of magnitude leaves the current above threshold unchanged, while the current below threshold scales to zero with the seed: the current is spontaneous. Full DMRG of the electron–photon problem (symbols) and the exact treatment of the photon in the adiabatic Hamiltonian (3) with the DMRG $E(\varphi)$ (lines) agree within a few percent. Beyond the maximum the current decreases as $I \simeq \hbar\omega\varphi^\star/2g^2$, because the condensate flux saturates at $\Phi_0/2$ while the effective inductance $\propto g^2/\hbar\omega$ grows.

Figure 5(b) shows the current in the (U, g) plane at fixed $\hbar\omega = 7.6\times10^{-4}t$. Its boundary has two parts. The lower edge, $U_c \simeq 0.9t$, is the 0–π transition of the junction itself. For $U < U_c$ the ground state is a singlet with $E_2 < 0$, the orbital response is diamagnetic, and no current appears at any g. The left edge, $g_c(U)$, follows the classical criterion $\beta(U, g) = 1$ and moves to stronger coupling as $E_2(U)$ decreases. Panel (c) is the same map from the mean-field theory of Ref. [1]: Hartree–Fock BdG electrons times a photon product state, with the same seed. For $U \lesssim 5t$, i.e. $E_J/\hbar\omega \gtrsim 3$, mean field reproduces U_c and $g_c(U)$ to within the grid resolution ($\delta g = 0.01$). It underestimates the maximal current by 10–20% (the product state renormalizes the junction bond by $|\langle D\rangle| < 1$). For $U \gtrsim 6t$, where E_2 drops to $E_J \lesssim 3\hbar\omega$, mean field predicts an ordered phase (dashed line in (b)) that is absent in the exact solution. Quantum fluctuations of the flux restore the symmetry there, and the ground state remains the cat state described below. In short, the product state is reliable only when $E_J \gg \hbar\omega$ and $g \ll 1$. Both conditions are violated by the parameters of the original notebook, see Fig. 6: at $\hbar\omega = 0.02$ the exact ground state carries no current at any g, whereas at $\hbar\omega = 0.002$ the same microscopic model orders at $g_c \simeq 0.21$, while mean field predicts $g_c \simeq 0.30$.

Breakdown of the product-state description for $E_J \simeq \hbar\omega$. We first take the parameters of a Zeeman-polarized dot embedded in a short BCS ring ($t = 2$, $\Delta = 0.6$, $L = 11$, $\epsilon_{d\uparrow} = -0.3$, $\epsilon_{d\downarrow} = 3.7$, $t_J = 0.5$, $\hbar\omega = 0.02$). The self-consistent product state predicts a current-carrying phase for $g \gtrsim 0.75$, with $\langle a^\dagger a\rangle \simeq 1$–$1.4$ and $|\langle I\rangle| \approx 0.02$ [Fig. 8]. DMRG shows that this phase is an artefact. The exact ground-state energy decreases monotonically with g and lies below the MF energy by up to $1.8\hbar\omega$ (at $g = 1.2$). The gap to the first excited state stays a sizeable fraction of $\hbar\omega$ ($E_1 - E_0 = 0.33\hbar\omega$ at $g = 1.2$), and the photon number saturates at $\langle a^\dagger a\rangle \simeq 0.36$. The response to the seed flux $\varphi_{\rm cl} = 10^{-3}$ is linear: $\langle I\rangle \approx 7\times10^{-5}$, more than two orders of magnitude below the MF value. Here and below we evaluate the current through the exact relation $\langle I\rangle = \hbar\omega\langle X\rangle/2g$, which follows from $\langle[H, P]\rangle = 0$ and states that the persistent current *is* the static displacement of the mode. DMRG and the adiabatic theory (3) agree to within $0.06\hbar\omega$ in energy and 1.3% in $E_1 - E_0$. The residual difference is the non-adiabatic (diagonal Born–Oppenheimer) correction, of relative order $\hbar\omega/E_A \approx 0.1$.

Two effects make the product state fail. (i) It is not gauge invariant [13, 14]: with the full Peierls phase on one bond, it renormalizes that bond by $|\langle D\rangle| < 1$ and so loses phase-*independent* kinetic energy, which the exact state recovers by following the flux adiabatically. (ii) Here $E_J \simeq \hbar\omega$. The Landau free energy does develop a double well for $g > 0.58$, but its depth is below the zero-point energy, and the flux distribution $P(X)$ stays essentially unimodal. A further subtlety concerns ring size: since $L \lesssim \xi$, $E(\varphi)$ of this ring is dominated by the single-electron harmonic, $E_1 = 2.1\times10^{-2} \gg E_2 = 2.0\times10^{-3}$ [Fig. 7(a)]. The instability would then be one of the normal h/e persistent current rather than of the Josephson junction. To isolate Josephson physics we use rings with $L \gg \xi$ below ($t = 1$, $\Delta = 1$, $L = 14$, giving $|E_1/E_2| \lesssim 10^{-2}$).

Half-flux-biased junction. For an ordinary weak link ($t_J = 0.5$, $E_2 = -0.056\, t$, Andreev gap $0.77\, t$) biased by $\Phi_0/2$, Fig. 9 shows the full DMRG solution at $E_J/\hbar\omega = 10$ as a function of β. The current rises steeply at $\beta \simeq 1.1$ and saturates at $|\langle I\rangle| \simeq 0.97 I_c$ for $\beta \approx 1.7$. It then decreases as the flux approaches $\Phi_0/2$, where the junction current vanishes. DMRG and the adiabatic theory agree to $0.003\hbar\omega$ in energy, to $\lesssim 5\%$ in current and to 2% in $E_1 - E_0$ at all β. The product state is 0.2–$0.4\hbar\omega$ too high in energy and overestimates the current threefold near threshold ($0.71 I_c$ versus $0.21 I_c$ at $\beta = 1.13$). It becomes accurate only deep in the ordered phase, where the photon is a nearly coherent state. The flux distribution extracted from the photon reduced density matrix evolves from a Gaussian centred at zero to a single displaced peak at $\pm\Phi^\star$. The sign is fixed by the residual odd harmonic $E_1 = -5.4\times10^{-4}$, which tilts the double well by $2|E_1|\sin\varphi^\star \approx 10^{-3}t \approx 0.2\hbar\omega$. This tilt sets the splitting $E_1 - E_0$ for $\beta > 1.2$ in DMRG and BO alike. Without it, $\delta/\hbar\omega$ would drop below 10^{-6} at $\beta = 2$. The external-bias route thus provides a current-carrying state whose direction is *selected* rather than spontaneously chosen. Its residual symmetry breaking is $\propto e^{-L/\xi}$ for a homogeneous ring, but in practice it is limited by the flux offset $\delta\Phi$.

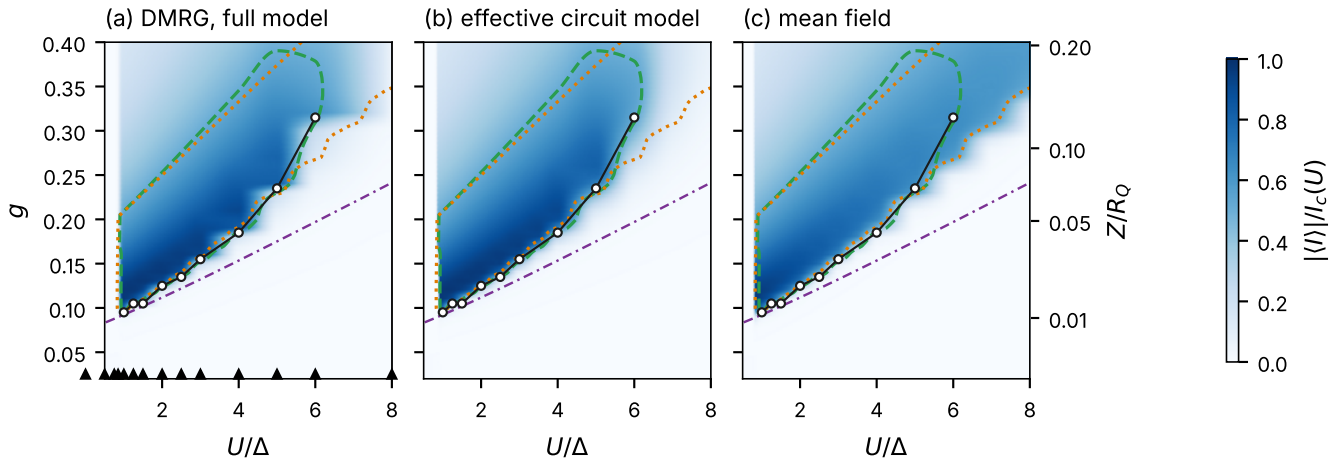


FIG. 2. Phase diagram of the microscopic model (1) in the plane of the dot charging energy U/Δ and the mode coupling g ($Z/R_Q = 4g^2/\pi$, right axis), at $\hbar\omega = 7.39 \times 10^{-4}\Delta$, $t_J = 0.5\Delta$, $L = 10$, no external flux (seed $\varphi_{\rm cl} = 10^{-3}$). Colour: persistent current $|\langle I\rangle|$ from the gauge-invariant operator (5) in units of the critical current $I_c(U)$ of the π junction. (a) Full DMRG of electrons and photon, both parity sectors (triangles: computed U values; g step 0.01). (b) Effective circuit model $H_{\rm eff}$ with the DMRG $E_s(\varphi; U)$. (c) Mean field. Circles: DMRG threshold ($|\langle I\rangle| = 0.5 I_c$, bar: grid step); green dashed and orange dotted: the same contour from the effective model and from mean field; purple dash-dotted: classical threshold $\beta = 1$.

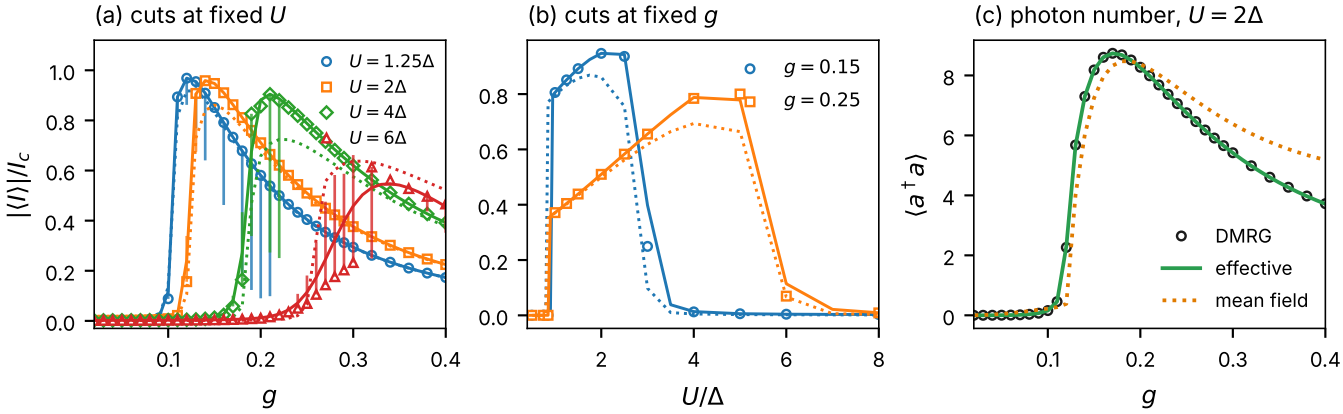


FIG. 3. Cuts through Fig. 2. (a) Current versus g at $U/\Delta = 1.25, 2, 4, 6$ and (b) versus U at $g = 0.15, 0.25$: full DMRG (symbols; vertical bars: bracket between the two initial states, converted to a current with Eq. (6)), effective circuit model (solid), mean field (dotted). (c) Photon number at $U = 2\Delta$.

Interaction-driven π junction without external flux. We now replace the weak link by an Anderson impurity at particle–hole symmetry ($\epsilon_d = -U/2$, $t_J = 0.5$) and set $\varphi_{\rm cl} = 0$. DMRG in the even and odd fermion-parity sectors [Fig. 7(b,c)] gives the 0–π transition known for magnetic quantum dots [15–21]. For $U < U_c \simeq 0.9\,t$ the singlet (even) state is the ground state at $\varphi = 0$ and the junction is a 0-junction. For $U > U_c$ the Kramers doublet (odd sector) is the ground state at all phases, and $E(\varphi)$ has its *minimum* at the Cooper-pair phase π. At $U = 2t$ we find $E_2 = +7.39 \times 10^{-3} t$, with $|E_1|, |E_4| < 10^{-4} t$: a pure π junction, indistinguishable after normalization from the half-flux-biased weak link [Fig. 7(a)]. The π coupling decreases monotonically with U ($E_2 \propto U^{-1.3}$ for $4t \le U \le 10t$), as expected for cotunnelling through the singly occupied level.

Coupling this junction to the mode [Fig. 10] gives the transition without any static flux. Here $H_{\rm ring}$ is real and $\langle I\rangle = 0$ in every eigenstate, so the order is diagnosed by the two lowest states. The splitting $\delta = E_1 - E_0$ collapses and the flux operator projected onto the quasi-degenerate doublet acquires the eigenvalues $\pm\Phi^\star$. At $E_J/\hbar\omega = 10$, DMRG and the adiabatic theory give the same $\Phi^\star$ to within 3% for all β [e.g. $\Phi^\star = 0.57\,\Phi_0/2$ at $\beta = 2$]. They also agree on $\delta/\hbar\omega = 0.71$ and 0.28 at $\beta = 0.5$ and 1. Deeper in the ordered phase, δ drops below the DMRG resolution for excited-state energies ($\sim 3 \times 10^{-5} t \approx 0.04\hbar\omega$, estimated by comparing $\chi = 256$ and 400 at $\beta = 1.5$). The adiabatic theory, validated above, then gives $\delta/\hbar\omega = 7 \times 10^{-4}$, 2×10^{-6}

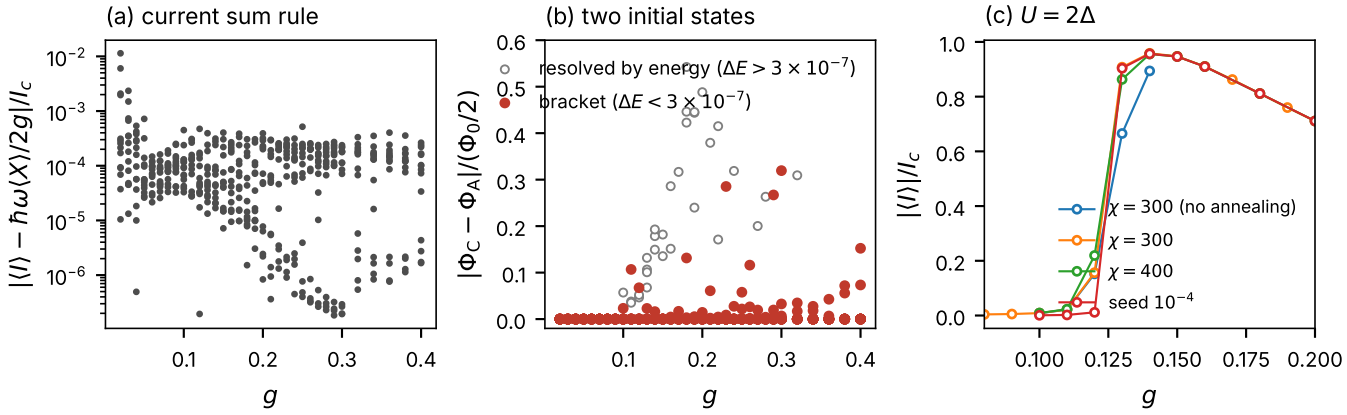


FIG. 4. Convergence of the full DMRG. (a) Violation of the sum rule (6) over the whole grid. (b) Difference between the fluxes of the two initial states (vacuum side A and polarized side C) in units of $\Phi_0/2$: open symbols, resolved by the energy; filled, reported as a bracket. (c) $U = 2\Delta$ near threshold: $\chi = 300$ without and with the annealed start, $\chi = 400$, and seed 10^{-4}.

and 2×10^{-8} at $\beta = 1.5$, 2 and 3. In the photon-number basis the DMRG ground state is a cat state: $P(\Phi)$ is bimodal with peaks at $\pm\Phi^\star$, whose separation tracks the classical minimum. With increasing $E_J/\hbar\omega$ the crossover sharpens into a pitchfork at $\beta = 1$ and δ vanishes exponentially [Fig. 10(b)]. The residual δ is the macroscopic tunnelling of a persistent current. Because it is protected by time-reversal symmetry, it is the only energy scale separating the two current directions. Each current state is additionally Kramers degenerate in the spin of the dot. The low-energy manifold is therefore a fourfold multiplet $\{\sigma = \uparrow,\downarrow\} \otimes \{\circlearrowleft,\circlearrowright\}$, split only by δ.

Experimental scales. The two control parameters are $\beta = 2\pi LI_c/\Phi_0$ and $E_J/\hbar\omega$, related by $\beta = 2\pi(Z/R_Q)(E_J/\hbar\omega)$, with $E_J/h \simeq 0.5\,\mathrm{GHz}\times I_c[\mathrm{nA}]$. Three classes of π elements span the phase diagram [Fig. 1(b)]: (i) SFS junctions ($I_c \sim$ 1–100 μA) reach $\beta > 1$ with geometric inductances of a few to a few hundred pH and $E_J/\hbar\omega \sim 10^2$–$10^4$. This is the classical limit, where the half-flux state is fully polarized, consistent with the spontaneous currents observed in SFS and d-wave π loops [5, 22, 23]. (ii) Quantum-dot π junctions in the doublet regime ($I_c^\pi \sim 0.1$–10 nA) require $L \sim 0.03$–3 μH, i.e. superinductances (Josephson-junction arrays, granular Al) as in fluxonium circuits [11, 12]; this is the π-qubit regime of Refs. [24–26]. These give $E_J/\hbar\omega \sim 1$: the ground state is the cat state $(|\circlearrowleft\rangle + |\circlearrowright\rangle)/\sqrt{2}$ with $\delta \sim 0.1\hbar\omega$, and the transition appears as a dip in the $|0\rangle \to |1\rangle$ frequency and a jump in the transition matrix element $\langle 0|\Phi|1\rangle$, accessible in two-tone spectroscopy. (iii) Multichannel or gate-tunable π junctions with $I_c \sim$ 50–200 nA in a high-kinetic-inductance loop ($L \sim$ 2–10 nH, $\omega/2\pi \sim$ 2–5 GHz) give $\beta \sim$ 1–3 and $E_J/\hbar\omega \sim$ 10–30. This is the regime of Figs. 9 and 10: a well-defined current-carrying state whose residual tunnelling $\delta/\hbar\omega \sim 10^{-4}$–$10^{-8}$ remains a measurable macroscopic-quantum-tunnelling signature. In the externally biased realization, the half-flux bias must be stable to $|\delta\Phi| \ll \delta/2I^\star$, i.e. $\lesssim 10^{-6}\Phi_0$ for the parameters above. The interaction-driven realization has no such requirement.

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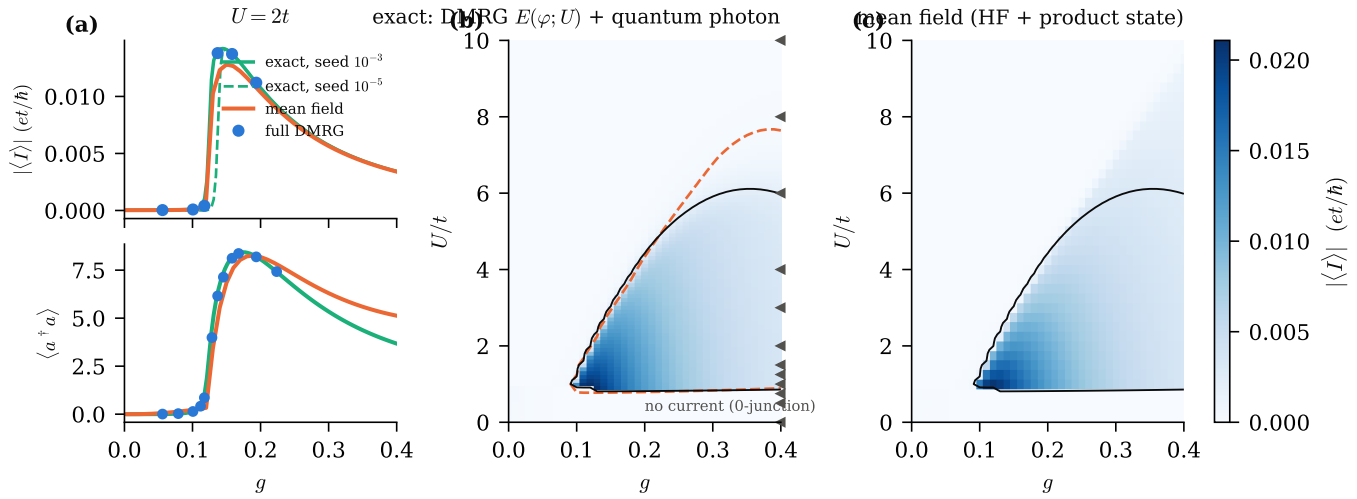


FIG. 5. Spontaneous persistent current in the microscopic model Eq. (1) ($L = 14$, $t = 1$, $\Delta = t$, $t_J = 0.5t$, $\epsilon_d = -U/2$, $\hbar\omega = 7.6 \times 10^{-4}t$, no external flux). (a) Current and photon number versus g at $U = 2t$: full DMRG (circles), exact photon in the adiabatic Hamiltonian with DMRG $E(\varphi)$ (green; solid and dashed: seeds $\varphi_{\rm cl} = 10^{-3}$ and 10^{-5}), and mean field (orange). (b) Current in the (U, g) plane (seed 10^{-3}), obtained in two steps: DMRG ground-state energy $E(\varphi; U)$ of the interacting ring at fixed phase (both parity sectors, U values marked by arrows, harmonics interpolated in between), then the photon treated exactly (full Fock space, no product ansatz) in $H_{\rm BO} = \hbar\omega a^\dagger a + E(\varphi_{\rm cl} + g(a + a^\dagger); U)$, Eq. (3); the only approximation is adiabaticity, $\hbar\omega/E_A \sim 10^{-3}$, checked against full DMRG in (a); black line: $|\langle I\rangle| = 2 \times 10^{-3} et/\hbar$; orange dashed: the same contour from mean field; arrows: U values computed by DMRG. (c) Mean field (Hartree–Fock BdG × photon product state), same color scale.

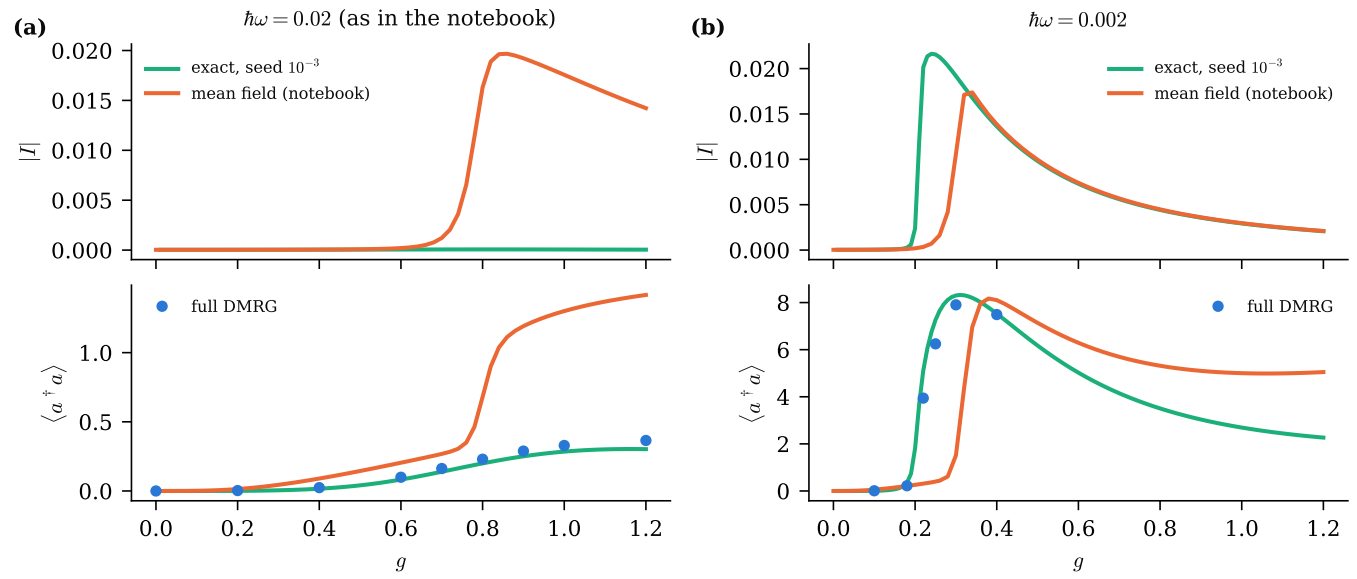


FIG. 6. Model of the original notebook (Zeeman-polarized dot, $L = 11$, $t = 2$, $\Delta = 0.6$): current (top) and photon number (bottom) versus g for (a) $\hbar\omega = 0.02$ and (b) $\hbar\omega = 0.002$. Green: exact (adiabatic, seed 10^{-3}); orange: mean field; circles: full DMRG.

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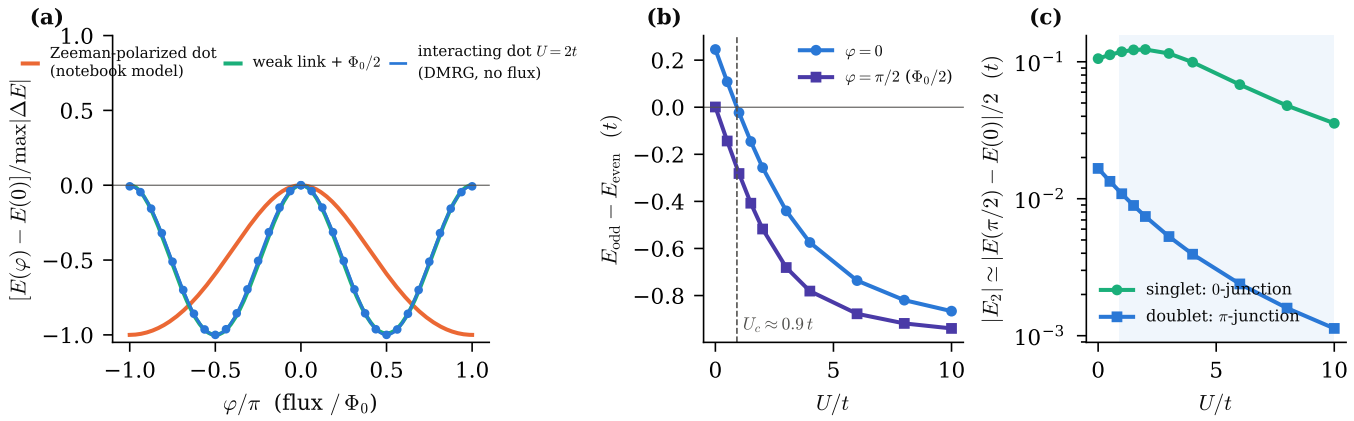


FIG. 7. (a) Normalized ground-state energy of the loop versus the electron phase φ (flux in units of $h/2e$) for three junctions: Zeeman-polarized dot in a short ring (dominated by the single-electron harmonic $\cos\varphi$), an ordinary weak link biased by $\Phi_0/2$, and an interacting dot at $U=2t$ without any bias (DMRG, symbols; line: cosine fit). (b) Parity competition $E_{\rm odd}-E_{\rm even}$ of the interacting-dot ring at $\varphi=0$ and $\varphi=\pi/2$ (DMRG, $L=14$, $\Delta=t$, $t_J=0.5t$). (c) Josephson energy $|E_2|$ in the singlet (0) and doublet (π) sectors. Shading: doublet ground state.

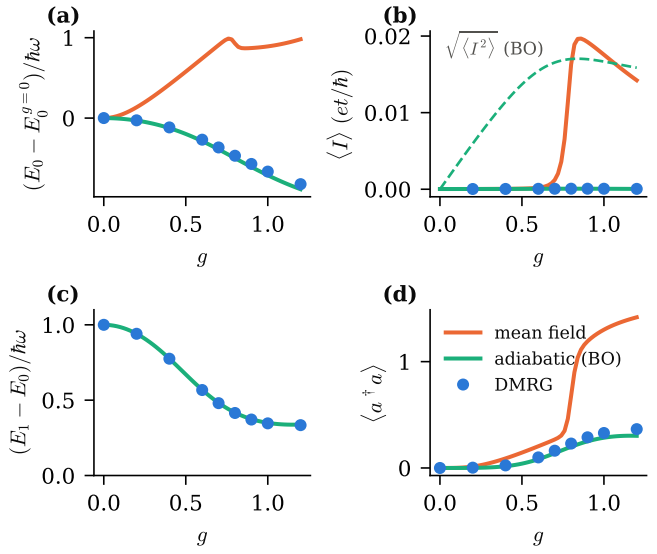


FIG. 8. Short ring with a Zeeman-polarized dot ($\hbar\omega=0.02$, $\varphi_{\rm cl}=10^{-3}$): mean field (MF, orange), adiabatic theory (3) (green), and DMRG (symbols). (a) Ground-state energy, (b) current (dashed: rms current of the BO ground state), (c) gap to the first excited state, (d) photon number.

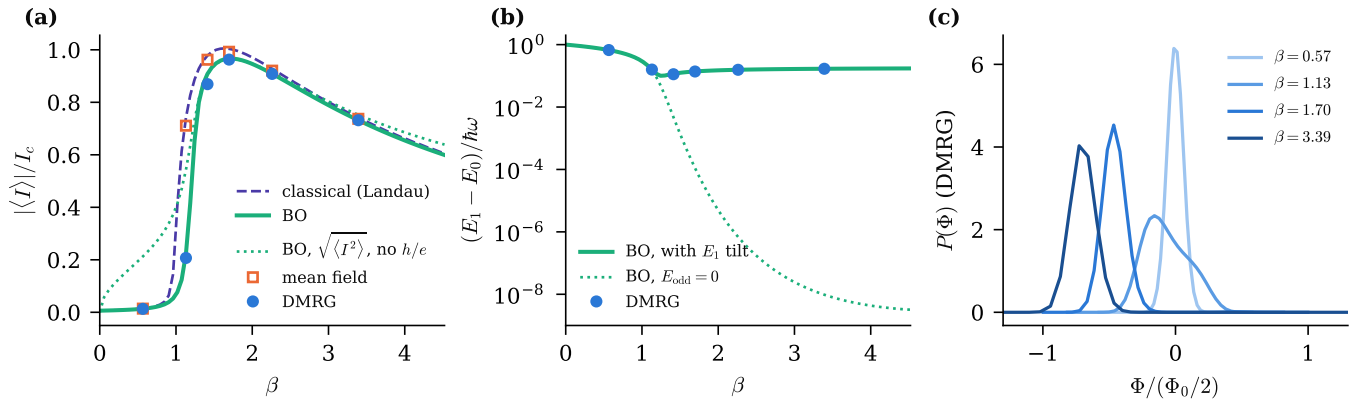


FIG. 9. Weak-link loop biased by $\Phi_0/2$, $E_J/\hbar\omega = 10$. (a) Persistent current versus β: DMRG (circles), MF (squares), adiabatic theory with (solid) and without (dotted) the odd harmonics, classical Landau theory (dashed). (b) Splitting of the two lowest levels. (c) Flux distribution of the DMRG ground state, obtained from the photon reduced density matrix.

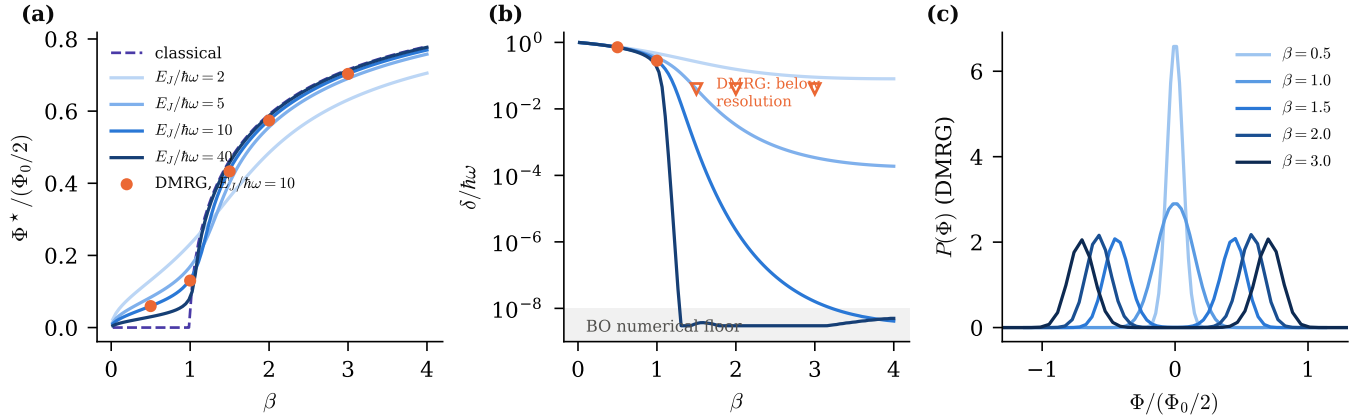


FIG. 10. Interaction-driven π loop ($U = 2t$, no external flux). (a) Flux of the condensate $\Phi^\star$ (eigenvalue of the flux projected on the two lowest states) versus β for increasing $E_J/\hbar\omega$ (lines: adiabatic theory with the DMRG $E(\varphi)$; circles: full DMRG at $E_J/\hbar\omega = 10$; dashed: classical). (b) Tunnel splitting δ between the two current-carrying states. (c) Flux distribution of the DMRG ground state (a cat state for $\beta > 1$).